

Problem 2.4

Find the amount of interest earned between time t and time n , where $t < n$, if

$$I_t = 2^t$$

for some positive integer n .

Hint: Recall the following sum from Calculus:

$$\sum_{i=0}^n ar^i = a \left(\frac{1 - r^{n+1}}{1 - r} \right), \quad r \neq 1.$$

Solution

The amount accumulated by time n is the sum of the interest amounts from time 0 up to time n :

$$A(n) = \sum_{i=0}^n 2^i.$$

Similarly, the amount accumulated by time t is

$$A(t) = \sum_{i=0}^t 2^i.$$

Therefore, the amount of interest earned between time t and time n is

$$A(n) - A(t) = \sum_{i=0}^n 2^i - \sum_{i=0}^t 2^i.$$

Now apply the geometric series formula with $a = 1$ and $r = 2$:

$$\sum_{i=0}^n 2^i = \frac{1 - 2^{n+1}}{1 - 2} = 2^{n+1} - 1,$$

and similarly,

$$\sum_{i=0}^t 2^i = \frac{1 - 2^{t+1}}{1 - 2} = 2^{t+1} - 1.$$

Substituting into the difference,

$$A(n) - A(t) = (2^{n+1} - 1) - (2^{t+1} - 1).$$

Thus,

$$A(n) - A(t) = 2^{n+1} - 2^{t+1}.$$

Hence, the amount of interest earned between time t and time n is

$$\boxed{2^{n+1} - 2^{t+1}}.$$